Higher Morita Categories

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1 Goal and Ideas

For a given enrichment context $\mathcal{V}$: $\mathbf{Alg}_{\mathcal{E}^n}(\mathbf{Pr}^L)$, we have a sequence of symmetric-monoidal functors

$$\mathbf{Alg}_{\mathcal{E}^n} \longrightarrow \mathbf{Mor}_n^\alpha \hookrightarrow \mathbf{Mor}_n^\beta \hookrightarrow \mathbf{Pr}_n^L \hookrightarrow \mathbf{Cat}_{(\infty,n)}^{\xrightarrow{\text{colim}}} \tag{1.1}$$

most of which are fully-faithful. The first category is $(\infty,1)$-$\mathcal{V}$-enriched, the others $(\infty,n+1)$-$\mathcal{V}$-enriched. The question of finding k-dualizable objects in these should be:

- (essentially) Kozsul duality in the first one (maybe more clear with augmented algebras),
- factorization homology in the Morita ones then smoothness/properness
- ...
- Construct a most general higher morita category by successive enrchiment. Requirement: stable under factorization homology.
- Embed such Morita category into nPr such that it controls all dualizable objects/cells.
- Duals up to n = adjoints, then duals at $n+1$ = finiteness, and duals at $> n+1$ = invertibility. Extend characterization to 1 β-level and recover **fusion** categories.
- Link with Kozsul duality? Which objects are part of a Kozsul pair?
- Link to K-theory?
- Compute factorization homology of Mor (which is the natural domain for.. factorization homology). This corresponds to finding its full pasting calculus.
- Compute Brauer spectrum for several enrichment context.

2 Notations and Assumptions

We assume a working theory of (∞, n)-categories, for example the iterated enrichment model, in which we have:

- for each $n \geq 0$, an $(\infty, 1)$-category $\mathbf{Cat}_{(\infty,n)}$,
- for each $n \geq 0$ and each $\mathcal{C} \colon \mathbf{Cat}_{(\infty,n)}$, a hom-functor
- $\mathcal{E}^k$-monoidal (∞, n)-categories $\mathcal{A}$ have a k-delooping $\mathbb{B}^k\mathcal{A}$ which is an $(\infty, n+k)$-category,
- the $(\infty, 1)$

3 Dualizable Objects

3.1 Adjunctions

$$\cdots \longrightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \longrightarrow \cdots \longrightarrow C_1 \xrightarrow{d_1} C_0 \longrightarrow 0$$

Definition 3.1 (Adjunction)**.** Let $\mathcal{C}$ be an $(\infty, 2)$-category. A morphism $f\colon a \to b\colon \mathcal{C}$ is a left (resp. right) adjoint if it is a left (resp. right) adjoint in the bicategory $\tau_{\leq 2}\mathcal{C}$. We then write $f \dashv g$ to denote that f is a left adjoint of g.

We have the following very powerful result by Riehl and Verity, essentially estabilishing that the higher homotopical uniqueness of the adjunction data:

Theorem 3.1 (Unicity of Adjoints)

There is a 2-category **Adj** *together with the data of two morphisms* $L, R\colon \mathbf{Adj}^{\to}$ *such that, for any* $\mathcal{C}$ $(\infty, 2)$*-category, the precompositions*

$$[\mathbf{Adj}, \mathcal{C}] \underset{R^*}{\overset{L^*}{\rightrightarrows}} \mathcal{C}^{\to}$$

induce ***monomorphisms*** *of spaces selecting the connected components of left (resp. right) adjoint morphisms.*

Geometrically, one can describe explicitly the 2-category **Adj** geometrically as follows:

- **Objects:** the positively and negatively oriented point $\{\pm\}$,
- **Morphisms:** finite 1-stratification of the real line where orientation of each segment alternates (we get sink and escape points in between them),
- **2-cells:** *(up to isotopy)* finite 2-stratification of the real plane where the oriented tangent vectors of the 1-stratas is not allowed to be $(-1, 0)$,

Combinatorially, one can also describe explicitly the hom-categories as:

$$\begin{cases} \mathbf{Adj}(+,+) = \Delta_+ = \mathrm{Free}^{\mathcal{E}^1}_{\mathcal{E}^1}(\{RL\}) \\ \mathbf{Adj}(+,-) = \Delta_\infty \\ \mathbf{Adj}(-,-) = \Delta_+^{\mathrm{op}} = \mathrm{Free}^{\text{co-}\mathcal{E}^1}_{\text{co-}\mathcal{E}^1}(\{LR\}) \\ \mathbf{Adj}(-,+) = \Delta_{-\infty} \end{cases}$$

where Δ_∞ (resp. $\Delta_{-\infty}$) is the category of finite non-empty linear orders with top (resp. bottom) element preserving morphisms.

Similarly, one can talk about k-morphisms in an (∞, n)-category being adjoints. Explicitly, if we let $\mathbf{Adj}_1 = \mathbf{Adj}$, one can build by induction the following "walking adjunct pair of k-morphisms" as the representing object in $\mathbf{Cat}_{(\infty,k+1)}$ of the functor sending $\mathcal{C}\colon \mathbf{Cat}_{(\infty,k+1)}$ to the fibration of spaces classifying

$$\begin{tikzcd} \mathbf{Cat}_{(\infty,k+1)}(\mathbf{Adj}_k, \mathcal{C}) \arrow[d, dashed] & & \\ \tau_0\mathcal{C} \times \tau_0\mathcal{C} \arrow[r, "\mathcal{C}(-,-)"'] & \mathbf{Cat}_{(\infty,k)} \arrow[r, "\mathbf{Cat}_{(\infty,k)}(\mathbf{Adj}_{k-1},-)"'] & \mathcal{S} \end{tikzcd}$$

We can also easily deduce unicity statements. More explicitly, for Θ_k the k-globe, we have $L, R\colon \mathbf{Adj}_k^{\Theta_k}$ k-cells, such that precomposition induces again monomorphisms

$$[\mathbf{Adj}_k, \mathcal{C}] \underset{R^*}{\overset{L^*}{\rightrightarrows}} \mathcal{C}^{\Theta_k}$$

3.2 Infinite Chain of Adjunctions

Definition 3.2 (Fully-Adjoint)**.** Let $\mathcal{C}$ be an $(\infty, 2)$-category and $f\colon a \to b$ a morphism. We say that f is part of an infinite chain of adjunctions if there exists a sequence of morphisms $f^{\vee n}$ such that:

$$\cdots \dashv f^{\vee -n} \dashv \cdots \dashv f^{\vee -1} \dashv f \dashv f^{\vee 1} \dashv \cdots \dashv f^{\vee n} \dashv \cdots$$

We say that f is infinitely-adjoint.

If we let $\mathcal{C}^{\infty\text{-adj}}$ be the following infinite fiber product,

$$\begin{tikzcd}
 & & & \mathcal{C}^{\infty\text{-adj}} \arrow[dll, hook] \arrow[dl, hook] \arrow[d, hook] \arrow[dr, hook] \arrow[drr, hook] & & \\
\cdots \arrow[dr, hook, "R"] & [\mathbf{Adj}, \mathcal{C}] \arrow[dl, hook, "L"'] \arrow[dr, hook, "R"] & [\mathbf{Adj}, \mathcal{C}] \arrow[dl, hook, "L"'] \arrow[dr, hook, "R"] & [\mathbf{Adj}, \mathcal{C}] \arrow[dl, hook, "L"'] \arrow[dr, hook, "R"] & \cdots \arrow[dl, hook, "L"'] \\
 & \mathcal{C}^{\to} & \mathcal{C}^{\to} & \mathcal{C}^{\to} & \mathcal{C}^{\to}
\end{tikzcd}$$

we can see that the projections down to any $\mathcal{C}^{\to}$ are monomorphisms of spaces selecting the connected components of infinitely-adjoint morphisms.

The functor $\mathcal{C} \mapsto \mathcal{C}^{\infty\text{-adj}}$ should then be representable by some 2-category that can be **geometrically** described as:

- **Objects:** the positively and negatively oriented point $\{\pm\}$,
- **Morphisms:**

Pablo: to be described

This is however beyond the scope of this document.

Similarly to the previous subsection, one can describe the same notion for k-morphisms in an (∞, n)-category of being infinitely-adjoint by taking the infinite fiber product

$$\begin{tikzcd}
 & & & \mathcal{C}^{\infty\text{-adj-}k\text{-cell}} \arrow[dll, hook] \arrow[dl, hook] \arrow[d, hook] \arrow[dr, hook] \arrow[drr, hook] & & \\
\cdots \arrow[dr, hook, "R"] & [\mathbf{Adj}_k, \mathcal{C}] \arrow[dl, hook, "L"'] \arrow[dr, hook, "R"] & [\mathbf{Adj}_k, \mathcal{C}] \arrow[dl, hook, "L"'] \arrow[dr, hook, "R"] & [\mathbf{Adj}_k, \mathcal{C}] \arrow[dl, hook, "L"'] \arrow[dr, hook, "R"] & \cdots \arrow[dl, hook, "L"'] \\
 & \mathcal{C}^{\Theta_k} & \mathcal{C}^{\Theta_k} & \mathcal{C}^{\Theta_k} & \mathcal{C}^{\Theta_k}
\end{tikzcd}$$

where importantly, once again, all projections are monomorphisms.

3.3 Iterated Adjunctions and Duals

Definition 3.3 (Fully-Adjoint)**.** Let $\mathcal{A}$ be an (∞, n)-category. By induction on $1 \leq l$, we define the following. For $1 \leq k$, a k-morphism $f\colon a_{k-1} \to b_{k-1}\colon \ldots : a_0 \to b_0$ is l-adjoint if:

- f is part of an infinite chain of adjunctions in $(\mathcal{A}(a_0, b_0)\ldots)(a_{k-2}, b_{k-2})$ (interpreted as $\mathcal{A}$ when $k = 1$)

  $$\cdots \dashv f^{\vee -n} \dashv \cdots \dashv f^{\vee -1} \dashv f \dashv f^{\vee 1} \dashv \cdots \dashv f^{\vee n} \dashv \cdots$$

  with (co)units $\epsilon(f^{\vee k})$, $\eta(f^{\vee k})$ of $f^{\vee k} \dashv f^{\vee k+1}$ for $k \in \mathbb{Z}$,
- (and if $l \leq 2$) for all $k \in \mathbb{Z}$, $\epsilon(f^{\vee k})$ and $\eta(f^{\vee k})$ are $(l-1)$-adjoint.

If $k + l = n$, we say that f is fully-adjoint.

Remark 3.1. For $n + 1 \geq k + l$, we recover the notion of invertibility.

Definition 3.4 (Fully-Dualizable Object)**.** Let $\mathcal{A}$ be an $\mathcal{E}^1$-monoïdal (∞, n)-category. For $1 \leq l$, an object $a\colon \mathcal{A}$ is l-dualizable if it is l-adjoint as a 1-morphism of $\mathbb{B}\mathcal{A}$. If $l = n$, we say that a is fully-dualizable.

Definition 3.5 (Has Adjoints or Duals)**.**

- Let $\mathcal{A}$ be an (∞, n)-category. We say that $\mathcal{A}$ **has adjoints** if all k-morphisms have a left and right adjoint for $1 \leq k \leq n-1$, or equivalently, if all k-morphisms are fully-adjoints.
- Let $\mathcal{A}$ be an $\mathcal{E}^1$-monoïdal (∞, n)-category. We say that $\mathcal{A}$ **has duals** if $\mathbb{B}\mathcal{A}$ has adjoints.

Remark 3.2. If $\mathcal{A}$ is $\mathcal{E}^k$-monoïdal for $k \geq 2$, then an object (seen as a morphism of $\mathbb{B}\mathcal{A}$) has a left adjoint if and only if it has a right adjoint if and only if it is part of an infinite chain of adjunctions. We get identifications $a^{\vee 2n} \simeq a$, which can be made canonical for $\mathcal{A}$ symmetric-monoïdal.

Example: $\mathbf{Bord}^{\mathrm{fr}}$ is symmetric-monoïdal and has duals. In particular, the point $\{+\}$ is a fully-dualizable object.

For the next result, we need to briefly define a notation for the data involved in a l-adjunction:

Definition 3.6 (Higher Adjunction Data)**.** An adjunction data is a word ω using the symbols ϵ, η, $-^{\vee n}$ (for any $n \in \mathbb{Z}$) applied to f. The depth of an adjunction data is the number $\mathrm{depth}(\omega)$ of times either of the letters η or ϵ were used.

For example $\eta(\eta(\epsilon(f^{\vee -5}))^{\vee 12})$ is an adjunction data of depth 3.

Theorem 3.2 (Unicity)

Let $\mathcal{A}$ be an (∞, n)-category, $1 \leq k, l$, $f\colon a_{k-1} \to b_{k-1}\colon \ldots : a_0 \to b_0$ a l-adjoint k-morphism l-adjoint and ω be a word Let ω be any word using the symbols ϵ, η, $-^{\vee n}$ (for any $n \in \mathbb{Z}$) applied to f.

3.4 Finding Fully-Dualizable Objects

Remark 3.3. Recall that we have functors $\mathbf{Bord}_k^{\mathrm{fr}} \to \mathbf{Bord}_n^{\mathrm{fr}}$ given by extending the framing trivially. **Warning:** these functors are **not** truncations.

Lemma 3.3 (Constructing Full-Adjunctions)

Let $\mathcal{A}$ be a symmetric-monoïdal $(\infty, n+1)$-category together with a functor $F\colon \mathbf{Bord}_n^{\mathrm{fr}} \to \mathcal{A}$ sending $\{+\}$ to some object a. Then A is fully-dualizable if and only if, for every n-morphism

$$f\colon a_{n-1} \to b_{n-1}\colon \ldots : a_0 \to b_0\colon \mathbf{Bord}_n^{\mathrm{fr}}$$

Ff is part of an infinite chain of adjunctions.

Proof: For $n = 0$, $\mathbf{Bord}_0^{\mathrm{fr}}$ is the trivial category $\{\bullet\}$ and the statement is tautological. Now assume it true for $n \geq 0$. Clearly, A (seen as a morphism in $\mathbb{B}\mathcal{A}$) is part of an infnite chain of adjunctions given by the image of $\mathbf{Bord}_1^{\mathrm{fr}} \to \mathbf{Bord}_n^{\mathrm{fr}} \to \mathcal{A}$. ■

4 Alpha-Case: Objects

Recall that $\mathbf{Mor}_n$ is an $(\infty, n+1)$-category.

4.1 First Step: n-dualizability

It's well-known that every k-morphisms of $\mathbf{Mor}_n$ is $(n-k)$-adjoints (but not necessarily fully-adjoint). Using factorization homology, we can make the n-dualizability data of an object more explicit:

Theorem 4.1 (Cite Claudia and John)

Let $\mathcal{V}\colon \mathbf{Alg}_{\mathcal{E}^n}\left(\mathbf{Cat}_{(\infty,1)}^{\text{sifted}-\text{colim}}\right)$ *. For* $a\colon \mathbf{Alg}_{\mathcal{E}^n}(\mathcal{V})$*, we have a functor*

$$\int_{-} a\colon \mathbf{Bord}_n^{\text{fr}} \longrightarrow \mathbf{Mor}_n(\mathcal{V})$$

from an (∞, n)*-category to an* $(\infty, n+1)$*-category.*

By Lemma 3.3, we then have the following:

4.2 Second Step: (n+1)-cells

For A to be $n+1$-dualizable, it suffices for the n-cells involved in its n-dualizable structure to have adjoints. A priori, the cobordism hypothesis's inductive proof relies on the claim that it's enough to check adjoints on some of these cells (the spheres). We can prove it formally in this context.

The strategy is the following:

(i) Any arbitrarily complicated n-cell $x\colon a_{n-1} \to b_{n-1}\colon \ldots \colon a_0 \to b_0$ in the image of our functor, can be seen (after smoothing) as the data of a n-manifold with boundary M and where the action is given by the canonical map of $\mathcal{E}^1$-algebra

$$\int_{\partial M \times \mathbb{R}} A \longrightarrow \operatorname{End}\left(\int_M A\right)$$

(ii) To find a dual, we need to

5 Alpha-Case: Higher Cells

Lots of elements in the proof are based on the idea that we have an inductive definition of $\mathbf{Mor}_n$, which was proved formally by Rune Haugseng I believe (Thm 1.3). For dualizability to be recognized as a finiteness condition, one simply need to find a type of ∞-categories left unchanged by this inductive process:

Definition 5.1 (Good Categories)**.** A n-good category is a stable presentable $\mathcal{D}\mathrm{isk}_n$-$\infty$-category, *i.e.* a n-disk algebra in the ∞-category of stable presentable categories with its usual symmetric tensor product.

Lemma 5.1 (Categories of Bimodules are Good)

Let $\mathcal{S}$ be a n-good category and let $a, b\colon \mathcal{D}\mathrm{isk}_n(\mathcal{S})$ be n-disk algebra objects. Then ${}_a\mathcal{B}\mathrm{imod}_b(\mathcal{S})$ is a $(n-1)$-good category.

Proof: Stability and presentability should probably be proven in [HA]. The $(n-1)$-disk operation should be clear to define. ■

We first need to identify which n-cells are dualizable. First, we need to define a higher notion of two-sided bar construction (arguably a "stratified-S^{n-1}"-sided bar construction).

Definition 5.2 (Higher Bar Construction)**.** Let us define a stratification of $\mathbb{R}^n$ as follows: for $0 \leq k \leq n-1$, we have two strata of the form $\{0\}^k \times \mathbb{R}_\epsilon \times \mathbb{R}^{n-k-1}$ with respectively $\epsilon \in \{>0, <0\}$ and they are respectively called A_k and B_k, and we have the additional strata $\{0\}^n$ called X. We can this stratified space Bar_n. For any $x\colon a_{n-1} \to b_{n-1}\colon \ldots \colon a_0 \to b_0$ in $\mathbf{Mor}_n(\mathcal{S})$, we can then define $\int_{\mathrm{Bar}_n} x$ where the stratas are labelled in the obvious way.

Lemma 5.2 (Dualizability of the highest-level cells)

A n-cell $x\colon a_{n-1} \to b_{n-1}\colon \ldots \colon a_0 \to b_0$ is dualizable in $\mathbf{Mor}_n(\mathcal{S})$ if and only if x is a finite $\int_{\mathrm{Bar}_{n-1}} a_{n-1}$-left-module and a finite $\int_{\mathrm{Bar}_{n-1}} b_{n-1}$-right-module in $\mathcal{S}$.

Proof: By the iterative definition of $\mathbf{Mor}_n(\mathcal{S})$, we see that x is dualizable whenever it is dualizable as a 1-cell in $\mathbf{Mor}_1({}_{a_{n-2}}\mathcal{B}\mathrm{imod}_{b_{n-2}}(\ldots {}_{a_0}\mathcal{B}\mathrm{imod}_{b_0}(\mathcal{S})))$. By [Insert reference to HA], x is then dualizable if and only if it is a finite a_{n-1}-left-module and a finite b_{n-1}-right-module in this iterated bimodule category. Equivalently, it has to be dualizable as a finite $\int_{\mathrm{Bar}_{n-1}} a_{n-1}$-left-module and a finite $\int_{\mathrm{Bar}_{n-1}} b_{n-1}$-right-module in $\mathcal{S}$ directly (the categories should be equivalent, would be interesting to prove carefully). ■

Theorem 5.3 (Dualizability in $\mathbf{Mor}_n$)

A k-cell $x\colon a_{k-1} \to b_{k-1}\colon \ldots \colon a_0 \to b_0$ is dualizable if and only if

- *x is a finite $\int_{\mathrm{Bar}_{k-1}} a_{k-1}$-left-module and a finite $\int_{\mathrm{Bar}_{k-1}} b_{k-1}$-right-module in $\mathcal{S}$,*
- *for all $0 \leq m \leq n-k-1$, x is a finite $\int_{X_m} x$-module where X_m is the stratified space $\mathbb{R}^{m+1} \times \mathrm{Bar}_{k-1}$ with an additional strata $S^m \subset \mathbb{R}^{m+1} \times \{0\}^{k-1}$ with x supported on the m-sphere, either a_{k-1} or b_{k-1} supported on the inside or outside (one can further multiply the space by $\mathbb{R}^{n-m-k}$ which is at least 1-dimensional, hence it gives an algebra).*

Proof: Dualizability can be checked at the level of 1-cells in $\mathbf{Mor}_{n-k+1}({}_{a_{k-2}}\mathcal{B}\mathrm{imod}_{b_{k-2}}(\ldots {}_{a_0}\mathcal{B}\mathrm{imod}_{b_0}(\mathcal{S})))$. ■

What does the frame version look like? One has to use that $S^m \times \mathbb{R}$ is canonically $(m+1)$-framed.